Main

ReplayTheatre.ai

Heckyeah2.8 (No. 11)

Git repo: https://github.com/SLab28/ReplayTheater.ai
FigJam:
https://www.figma.com/board/jPPrtDvKn1Hf9BQdMjIuk8/ReplayTheater.ai?t=iIKlatxaHKmJlf2x-1

Submission deadline: 29/02 2pm

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Tools
https://runware.ai/
https://platform.claude.com/docs/en/agent-sdk/overview

Roles

Visuals: WebXR interface and voice responsive agent + video generation life with stable diffusion

Interaction: Voice to text and text to story elements (json to be forwarded to SD)

Knowledge Base: ground truth about the Playback Theatre to inform the questions asked by the agent and the personality traits + guide rails

Judging criteria

25%
Future Potential
What is the project's long-term potential for success, growth and impact? How are they evaluating the quality and consistency of the outcome?

25%
Demo
How well has the team implemented the idea? Does it work?

25%
Creativity / Uniqueness
Is the project's concept innovative and unique?

25%
Pitching Quality
How well does the team present their project?

Ideation

- Someone to speak to? Social companionship with AR entity/agent
    - Affective API
    - Hospital companion for oncology department
    - Responsive sphere
    - Like Replika but with emotion recognition and abstracted
- Psychedelics therapy
- Replay Theater: tell the story to the actors, bring to live → stable diffusion 4
    - User journey:
        - Conversation
        - Follow up questions
        - Replay
        - Wrap and put away

- Exposure therapy by generating 3D worlds for someone's experience to experience in VR
    - Or, simulating social situations for people with social anxiety using LLMs (think Uncover the Smoking Gun for characters)
- Consuming psychiatric/psychotherapist scribe content to identify particularities in speech (spoken and text) and compare to existing cases for treatment outcomes
- Neuromorphic computing
- Keyboard app for smartphones that analysis changes in sentiment overtime to track relapses/crises and report to psych/caretaker, alert nurse to make a phonecall

Pipeline:
1. Problem Statement: Identify research, voice why this is important and **why now**
2. User Journey
3. System Architecture
4. Divide → Hack!

Problem Statement

Playback Theater:

Message Box: promise (context) → limitation → solution → impact

Project Description

Tech Stack

WebXR
Stable Diffusion
Python
THREE.js

Sponsor Tech

Claude API

Team

Sonia -
Moein -
Yasha -

Links

https://www.londonplayback.com/what-is-playback
https://fass.open.ac.uk/research/projects/pasar/videos/playback-theatre

Research

- A Pilot Study on Playback Theatre as a Therapeutic Aid after Natural Disasters: Brain Connectivity Mechanisms of Effects on Anxiety - PubMed
- Psychotherapeutic Playback Theatre, Well-Being, and Psychological Distress: An Impact Study

Landscape

Need

Need
What is the pressing need that we address?

Users

https://www.sciencedirect.com/science/article/pii/S0040162523003190v

What is the need? → (un)process, externalise/experience, bubble, trapped/ease, perceptual, relief

Companionship → growing rates of social isolation, people turning to digital agents looking for social fulfilment. Part of how we cope with the past experiences is through telling stories. Traditionally people would meet to exchange memories and past experiences. With many aspects of social interaction moving to the digital space there is a need for new ways of being able to cope with past experiences.

Playback Theater → offers a new perceptual way to relieve past experiences. It is a unique form of improvisational theatre in which audience members volunteer stories from their lives and see them played back on the spot [1]. Social aspect of the therapy.

Wrap and put away → physical act of letting go

Feelings we want to evoke: "Theraputic/Relieving, intimate and enjoyable"
Lifting up → reexperience → anew → replay

Among the different therapeutic elements that have been highlighted as key contributors to the success of the psychotherapeutic process, are the experiencing of both a strong emotional bond between the participants (usually patient, individually or in groups, and therapist) and a setting that is regarded as a location of cure, together with a set of procedures and/or rituals that induce the patient to extend something that is regarded as positive, useful and adaptive (Laska et al., 2014, Sousa, 2017), which commonly gets one to look differently at one's own circumstance, being it in a more cognitive way or an aesthetic one [2].

Interpretative Phenomenological Analysis (IPA)→ evaluation [3].

' psychological restoration and healing can only occur by making people feel heard and re-experience their own stories in a protected environment [5].

The increasing need for mental health support and a shortage of therapists have led to the development of the eXtended-reality Artificial Intelligence Assistant (XAIA) [4].

References

[1] https://www.londonplayback.com/
[2] https://www.sciencedirect.com/science/article/pii/S0197455624000376?via%3Dihub
[3] https://www.cureus.com/articles/357487-witnessing-embodying-and-connecting-a-phenomenological-study-of-playback-theatre#!/
[4] https://www.nature.com/articles/s41746-024-01011-0
[5] https://doi.org/10.70121/001c.123641

Market Research/Competitors

Link to Perplexity's analysis of market
Link to Claude's Research on the same topic

tl'dr below:

From Claude's

Every individual element of Replay Theater's concept has been independently validated. AI companions can form therapeutic alliances comparable to human therapists (Therabot RCT, NEJM AI, 2025). VR therapy produces clinically significant outcomes for PTSD, pain, anxiety, and psychosis (BraveMind, RelieVRx, gameChange — all with published RCTs). Personal memory cues in immersive environments facilitate reconsolidation (3MDR across four-country military trials). Generative AI can create personalized therapeutic imagery and environments (PsyTechVR MindGap, Champalimaud Foundation). What does not yet exist is a product integrating all four into a coherent consumer experience.
The competitive window is narrowing. XRHealth's acquisition spree is building the infrastructure to add AI companionship. XAIA has proven the AI-plus-VR therapy concept with published evidence. PsyTechVR holds a patent on AI-generated personalized VR therapy environments. Talkspace is building a proprietary mental health LLM. The most defensible path for Replay Theater would combine three elements: (1) early clinical evidence demonstrating safety and efficacy of memory-based immersive therapy, (2) a dual B2C/B2B2C distribution model (avoiding the pure-consumer trap that killed Woebot, Pi, and Mindstrong), and (3) a regulatory strategy that positions for FDA Breakthrough Device Designation while the agency actively develops its generative AI framework. The science supports it. The market wants it. The question is execution speed — before the converging incumbents close the gap from both sides.

Links to existing products:

- https://www.rendever.com/alcove-sanctuary
- https://xaia.health/en/xr
- https://www.xr.health/ (FDA approved)
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PRD

https://claude.ai/share/5f1e9103-a45e-451b-bd19-79c0727f5a32

PRD: Digital Playback Theatre — Hackathon Demo

v3 • Scope: <24h build • Goal: Working demo + judge-ready presentation • Team: 3–5 people

Vision

A web app that guides a user to share a personal story (text or voice), immerses them in a virtual theatre through WebXR and GLSL shader art, then plays back an AI-generated video of their story on stage — recreating the therapeutic "witness effect" of playback theatre. The GLSL/WebXR environment soothes the user into the experience; the AI video is the performance.

Core Architecture (two phases)

Phase 1 — The Theatre (GLSL + WebXR): User enters a WebXR virtual theatre rendered in-browser. GLSL shaders create the ambient environment: soft particles, warm shifting light, gentle organic motion. This is where the guided conductor prompts appear. The user tells their story while already immersed in an intimate, soothing space. The theatre is the ritual container — it mirrors the physical experience of sitting in a playback theatre venue before the performance begins.

Transition — The Hold: After story capture, the theatre dims. GLSL shaders shift to a meditative waiting state (slow breathing light, floating motes). A message appears: "Your story is being heard…" Behind the scenes: Claude API converts the user's story into a scene description + Stable Diffusion video prompt. The prompt is sent to Stable Video Diffusion (via Nvidia or HuggingFace remote GPU). Expected wait: 30–120 seconds depending on clip length and queue.

Phase 2 — The Performance (AI Video): The theatre lights come up. The generated video plays on the stage within the WebXR environment. The user watches an AI-generated artistic video of their personal narrative — the core therapeutic moment. After playback: soft reflection prompt ("How did that feel?") with option to replay or close.

User Flow (~5–10 min)

Landing page (calm, minimal, ambient audio) → Enter theatre (WebXR activates; user sees an intimate stage with warm lighting and shader-driven ambient visuals) → Guided prompts (conductor questions appear in the theatre space, one at a time; user responds via text or voice) → Hold (theatre dims, meditative shader state, video generating in background) → Performance (video plays on stage) → Reflection ("How did that feel?" — replay / close).

Requirements

Area	Must Have (demo day)	Nice to Have
Input	Text input with prewritten guided prompts (3–4 conductor-style questions). Speech-to-text via browser Web Speech API. Input happens inside the WebXR theatre environment.	Ambient music during input. Gentle text animation on prompts. Voice tone analysis to adapt pacing.
AI pipeline	User responses → Claude API generates structured scene description + optimised Stable Diffusion video prompt (emphasising character emotion, intimacy, artistic style) → prompt sent to Stable Video Diffusion. Nvidia Nemotron available as fallback LLM.	Multi-scene chaining: Claude breaks story into 2–3 scenes, each generating a separate clip, stitched sequentially.
Video gen	Stable Video Diffusion via Nvidia GPU credits or HuggingFace Inference. Target: one 10–20s artistic video clip per story. Style enforced via prompt engineering: watercolour/painterly aesthetic, character-focused, warm muted palette. Negative prompts suppress photorealism and spectacle.	Multiple clips stitched into a 30–60s sequence. Runware ($30 credits) as emergency fallback if SD pipeline fails.
GLSL shaders	Theatre ambience: soft floating particles, warm pulsing light, gentle organic motion. Hold state: meditative breathing light effect, slow colour shifts. Transitions: smooth dim/brighten between phases. All shaders reinforce intimacy — slow, organic, warm.	Reactive shaders responding to ambient audio. Per-emotion shader variations (rain for sadness, warm glow for comfort). Subtle vignette and film grain on video playback.
WebXR theatre	Immersive virtual theatre built with Three.js + WebXR API. Intimate small-venue feel: dark surrounds, warm stage lighting, soft textures. Video plays on a stage surface within the 3D scene. Works	Spatial audio. Subtle environmental details (fabric curtains, candle flicker). Seated-mode optimised for VR comfort.

	in browser (desktop + mobile fallback as 2D view), VR headset optional.	
Guided UX	Prewritten prompt sequence for emotional safety: grounding opener ("Take a breath"), story elicitation ("Tell me about a moment that stayed with you"), feeling check ("What did you feel?"), closing intention ("What would you like to let go of?").	Adaptive follow-up questions via Claude based on user responses.
Safety	Content moderation on input (flag crisis language, show helpline info). Style-locked prompts prevent violent/graphic video output. Disclaimer: "This is not therapy."	Therapist-facing mode with session summary export.
Frontend	Single-page app built with Claude Code. Landing page outside WebXR; theatre experience inside WebXR. Dark, warm UI. Transitions between steps are slow and deliberate. Mobile-responsive 2D fallback for non-VR.	PWA wrapper. Particle animation on landing page.

Tech Stack

Build tool: Claude Code. LLM: Claude API (primary), Nvidia Nemotron (fallback). Video gen: Stable Video Diffusion via Nvidia GPU credits or HuggingFace Inference (primary) | Runware as emergency fallback ($30 cap). Shaders: Custom GLSL for theatre ambience, hold state, and transitions. Immersive: Three.js + WebXR API. Voice: Browser Web Speech API. Hosting: Vercel. Audio: Royalty-free ambient loops (Pixabay/Freesound).

Latency Strategy

Video generation will take 30–120s. This is not a bug — it's designed into the experience. The user is already inside the WebXR theatre when generation begins. The hold state uses GLSL shaders to create a meditative, breathing-light environment with the message "Your story is being heard." This frames the wait as part of the ritual, not a loading screen. For the live demo: pre-generate a backup video so the presentation is never blocked by API latency.

What Impresses Judges

The two-phase experience: entering a shader-lit virtual theatre, then watching your own story come to life as AI-generated video on stage. The GLSL work signals craftsmanship. The WebXR signals ambition. The calm, guided UX signals therapeutic maturity. The research grounding (playback theatre evidence base) signals depth. Bonus: a team member tells a real (light) story live, the audience watches the theatre dim, then the video appears. Pre-generate this backup to guarantee it works.

Scope Cuts + Post-Hackathon Roadmap

Not for demo day: user accounts, persistence, therapist dashboard, community features, multi-language, scalability, multi-scene stitching. Mention in pitch: Prolific user study planned for post-hackathon validation. Doubleword batch inference for cost-efficient scaling across hundreds of users. Full AR/XR extensions beyond WebVR.

Timeline (parallel workstreams)

Hours	Stream A: AI Pipeline	Stream B: WebXR + GLSL	Stream C: UX + Frontend
0–2	Prompt engineering: conductor questions + Claude → scene description → SD video prompt chain. Test SD video output.	Scaffold Three.js + WebXR theatre scene. Basic geometry: stage, seating, dark surrounds.	Landing page design. Guided prompt flow wireframe. Claude Code project scaffold.
2–6	End-to-end pipeline: user text → Claude → SD video. Iterate on prompt quality and style consistency.	GLSL shader development: ambient particles, warm pulsing light, hold-state breathing effect.	Build guided input UI (text + Web Speech API). Integrate input into WebXR scene.
6–10	Optimise video prompt for emotional fidelity. Test with 5+ diverse story inputs. Set up fallback (pre-generated video).	Video playback on stage surface in WebXR. Dim/brighten transitions between phases. Dissolve effects.	Connect AI pipeline to frontend. Wire up full flow: input → hold → playback → reflection.
10–14	Generate backup demo video. Stress-test pipeline reliability.	Polish: theatre lighting, shader tuning, ambient audio integration, mobile 2D fallback.	UI polish: transitions, typography, spacing, disclaimer. End-to-end testing with real stories.

14–16	All streams: rehearse demo, prep presentation slides, assign speaking roles, test VR headset setup.		